FROM BOCHNER TO THE LICHNEROWICZ BOUND

Positive Ricci curvature controls a positive Laplace eigenvalue through one eigenfunction. The useful bridge is the Hessian trace: it retains information that would be lost by simply discarding a nonnegative term.

1. The eigenvalue problem

We use the nonnegative Laplacian throughout. Gradient, covariant Hessian and divergence integration are assumed; the precise spectral and Bochner inputs are recalled before their use.

Theorem 1.1 (Lichnerowicz)**.** *Let (M^m, g) be a closed connected smooth Riemannian manifold of dimension $m \geq 2$. Suppose that for every $x \in M$ and $v \in T_xM$,*

$$\mathrm{Ric}_g(v, v) \geq (m-1)g(v, v).$$

For the nonnegative Laplacian $\Delta = -\operatorname{div}_g \nabla$, let λ_1 denote its smallest positive eigenvalue on smooth real functions. Then $\lambda_1 \geq m$. This classical bound makes no equality-rigidity claim.

The callable inputs. The following classical statements are imported, not proved here. They are the exact versions required by this argument [1, Sections 29.1–29.2].

Proposition 1.2. *Let (M^m, g) be a closed connected smooth Riemannian manifold with $m \geq 2$ and put $\Delta = -\operatorname{div}_g \nabla$. The scalar Laplacian has a smallest positive eigenvalue $\lambda_1 > 0$ with a nonzero smooth real eigenfunction f satisfying $\Delta f = \lambda_1 f$. For every smooth real h on M, the scalar Bochner identity in this sign convention is*

$$-\tfrac{1}{2}\Delta|\nabla h|^2 = |\operatorname{Hess} h|^2 - \langle \nabla \Delta h, \nabla h\rangle + \mathrm{Ric}(\nabla h, \nabla h).$$

Here $\operatorname{Hess} h(X, Y) = \langle \nabla_X \nabla h, Y\rangle$, its norm is the pointwise Hilbert–Schmidt norm, and $\operatorname{tr}_g \operatorname{Hess} h = -\Delta h$. The spectral existence and local Bochner identity are imported classical inputs.

The source uses $L = \operatorname{div} \nabla$. Replacing L by $-\Delta$ accounts for both minus signs in the displayed identity. No equality case or classification theorem is included among the imports.

Integrating the identity. For smooth h, k, the product rule and closedness give

$$0 = \int_M \operatorname{div}(h \nabla k)\, dV_g = \int_M \big(\langle \nabla h, \nabla k\rangle - h\Delta k\big)\, dV_g.$$

In particular, $\int_M \Delta q\, dV_g = 0$ for smooth q. These formulas use the Riemannian volume density and do not require an orientation.

Choose a nonzero first eigenfunction f and write $\lambda = \lambda_1$. Substitution in Bochner and integration yield

$$\lambda \int_M |\nabla f|^2 = \int_M \big(|\operatorname{Hess} f|^2 + \mathrm{Ric}(\nabla f, \nabla f)\big). \tag{1.1}$$

Here and below the measure is dV_g. The same integration by parts gives

$$\int_M |\nabla f|^2 = \int_M f\Delta f = \lambda \int_M f^2. \tag{1.2}$$

Retaining the trace. At each point choose a g-orthonormal frame and put $H_{ab} = \operatorname{Hess} f(e_a, e_b)$. Cauchy–Schwarz on its diagonal entries gives

$$(\operatorname{tr} H)^2 = \left(\sum_a H_{aa}\right)^2 \le m \sum_a H_{aa}^2 \le m \sum_{a,b} H_{ab}^2 = m|H|^2.$$

Since $\operatorname{tr} H = -\Delta f$, this supplies exactly the term needed in (1.1):

$$|\operatorname{Hess} f|^2 \ge \frac{(\Delta f)^2}{m} = \frac{\lambda^2 f^2}{m}. \tag{1.3}$$

Closing the connection. Let $I = \int_M f^2 > 0$. Insert the curvature lower bound and (1.3) into (1.1), then use (1.2):

$$\lambda^2 I \ge \frac{\lambda^2}{m} I + (m-1)\lambda I.$$

Thus $\frac{m-1}{m}\lambda^2 I \ge (m-1)\lambda I$. Each factor in $(m-1)\lambda I$ is positive: $m \ge 2$, $\lambda > 0$, and $f \ne 0$. Division gives $\lambda \ge m$, proving Theorem 1.1.

2. Why these inputs cannot be dropped

Discarding $|\operatorname{Hess} f|^2$ in (1.1) would give only $\lambda \ge m - 1$. Its trace restores the stronger bound, which explains why Cauchy–Schwarz is a genuine point of the proof.

Dimension one is a true boundary. On the circle of radius R, Ricci curvature is zero, but its first positive eigenvalue is R^{-2}:

$$\Delta \cos\theta = R^{-2} \cos\theta.$$

For $R > 1$ this is less than 1. Correspondingly, the factor $m - 1$ in the attempted division would vanish.

On a manifold with boundary, the divergence integral acquires a boundary term. This proof therefore establishes no boundary eigenvalue theorem. Likewise, equality in the displayed inequalities is not itself a proof of geometric rigidity.

References

[1] Ved Datar. Lectures on riemannian geometry. Online source, 2025. unpublished; controlling version: September 11, 2025.